

FUNCTIONAL REQUIREMENT DOCUMENT

# AssetFlow

Enterprise Asset & Resource Management System — Odoo Implementation

|               |                                       |
|---------------|---------------------------------------|
| Document Type | Functional Requirement Document (FRD) |
| Platform      | Odoo ERP                              |
| Prepared For  | Development Team (4 Members)          |
| Prepared By   | ERP Functional Consultant             |
| Status        | Implementation-Ready                  |

# 1. Project Overview

AssetFlow is an Odoo-based Enterprise Asset & Resource Management System that centralizes the registration, allocation, booking, maintenance, and audit of physical assets and shared resources across departments. The system enforces role-based access (Admin, Asset Manager, Department Head, Employee), prevents double-allocation and booking overlaps, and drives structured approval workflows for transfers, maintenance, and audits. It excludes purchasing, invoicing, and accounting functionality.

## 2. Functional Modules

### Module 1 — Authentication & Access

#### Objective

Provide secure, realistic account creation and login without self-elevated privileges.

#### Features

- Employee self-signup (creates Employee role only — no role selection at signup)
- Email + password login
- Forgot password / reset password flow
- Session validation and auto-logout on token expiry
- Role promotion restricted to Admin via Organization Setup → Employee Directory (Module 3)

#### Forms

- Signup Form
- Login Form
- Forgot Password Form
- Reset Password Form

#### Fields

Full Name, Work Email (unique), Password, Confirm Password, Department (optional at signup), Status (Active/Inactive — default Active).

#### Buttons

Sign Up, Login, Forgot Password, Send Reset Link, Reset Password, Logout.

#### Business Rules

- Every new signup is created strictly as Employee; role field is never exposed on the signup form
- Only Admin can promote a user to Department Head or Asset Manager (from Module 3, Tab C)
- A user cannot self-assign or self-elevate any role

#### Validations

- Email must be unique and valid format
- Password minimum complexity (length, mix of characters)
- Inactive accounts cannot log in

#### Workflow

Signup → Employee account created (role = Employee) → Login → Session validated → Dashboard (Module 2). Role change only via Admin action in Module 3.

#### User Roles

All roles use this module; only Admin can change another user's role.

#### Dependencies

None (foundation module). Employee Directory (Module 3) depends on this module's account records.

### Module 2 — Dashboard (KPI Home)

#### Objective

Give every role a real-time, role-scoped operational snapshot on login.

#### Features

- KPI cards: Assets Available, Assets Allocated, Maintenance Today, Active Bookings, Pending Transfers, Upcoming Returns
- Overdue Returns list shown separately from Upcoming Returns, visually highlighted
- Quick action shortcuts: Register Asset, Book Resource, Raise Maintenance Request
- Role-based widget visibility (e.g. Employee sees only their own allocations/bookings; Admin sees organization-wide data)

#### Forms

None (read-only dashboard with quick-action shortcuts that open Module 4/6/7 forms).

#### Fields

KPI counters (computed, not stored): available\_count, allocated\_count, maintenance\_today\_count, active\_bookings\_count, pending\_transfers\_count, upcoming\_returns\_count, overdue\_returns\_count.

#### Buttons

Register Asset, Book Resource, Raise Maintenance Request, View All (per KPI card).

#### Business Rules

- Overdue = Expected Return Date < today AND asset/booking still not returned/closed
- KPI values recompute on every dashboard load (or via scheduled cache refresh)
- Employees only see data scoped to themselves; Department Heads scoped to their department; Admin/Asset Manager see all

#### Validations

N/A (aggregation/read screen).

#### Workflow

Login → Dashboard loads → KPIs computed per role scope → user drills into a KPI or uses a quick action.

User Roles

Admin, Asset Manager, Department Head, Employee — all with scoped visibility.

Dependencies

Depends on Modules 4 (Assets), 5 (Allocation), 6 (Booking), 7 (Maintenance) for data aggregation.

Module 3 — Organization Setup

Objective

Maintain master data (departments, categories, employees/roles) that all other modules depend on. Admin-only.

Features

- **Tab A - Department Management:** create/edit/deactivate department; assign Department Head; optional Parent Department for hierarchy; Status
- **Tab B - Asset Category Management:** create/edit categories (Electronics, Furniture, Vehicles, etc.); optional category-specific fields (e.g. warranty period)
- **Tab C - Employee Directory:** view/search employees; Name, Email, Department, Role, Status; promote Employee → Department Head / Asset Manager (only place roles are assigned)

Forms

Department Form, Asset Category Form, Employee Directory List + Role Assignment Dialog.

Fields

- **Department:** Name, Code, Parent Department, Department Head (M2O Employee), Status
- **Asset Category:** Name, Code, Parent Category (optional), Custom Field Definitions (JSON/config), Status
- **Employee Directory:** Name, Email, Department, Role (Employee/Department Head/Asset Manager/Admin), Status

Buttons

New Department, New Category, Deactivate, Promote to Department Head, Promote to Asset Manager, Revoke Role, Save, Cancel.

Business Rules

- Only Admin can access this screen and all three tabs
- Only one active Department Head per department (configurable)
- Deactivating a department does not delete historical asset/allocation records
- A category cannot be deleted if assets are linked to it (deactivate instead)

Validations

- Department Name unique per company
- Category Name unique
- Parent Department cannot create a circular hierarchy

Workflow

Admin logs in → Organization Setup → creates Departments/Categories → opens Employee Directory → selects Employee → Promote to Department Head/Asset Manager → role updated → user gains new permissions on next login/session refresh.

User Roles

Admin only (full CRUD). All other roles: no access.

Dependencies

Depends on Module 1 (Employee accounts must exist). Feeds Modules 4, 5, 6, 7, 8.

Module 4 — Asset Registration & Directory

Objective

Register assets and provide centralized search/tracking with full lifecycle visibility.

Features

- Register asset with auto-generated Asset Tag (e.g. AF-0001)
- Search/filter by Asset Tag, Serial Number, QR code, category, status, department, location
- Lifecycle status display: Available, Allocated, Reserved, Under Maintenance, Lost, Retired, Disposed
- Per-asset history tabs: Allocation History, Maintenance History
- QR code generation per asset for scan-based lookup
- Mark asset as "shared/bookable" for Module 6

Forms

Asset Registration Form, Asset Search/Filter Panel, Asset Detail View (with History tabs).

Fields

Asset Tag (auto, unique), Name, Category (M2O), Serial Number, Acquisition Date, Acquisition Cost (report-only, not linked to accounting), Condition (New/Good/Fair/Poor/Damaged), Location, Photo/Documents (attachments), Shared/Bookable flag (boolean), Status (computed/lifecycle), QR Code (auto-generated), Current Holder (computed).

Buttons

Register Asset, Save, Generate QR, Scan QR, Edit, Retire Asset, Dispose Asset, View History.

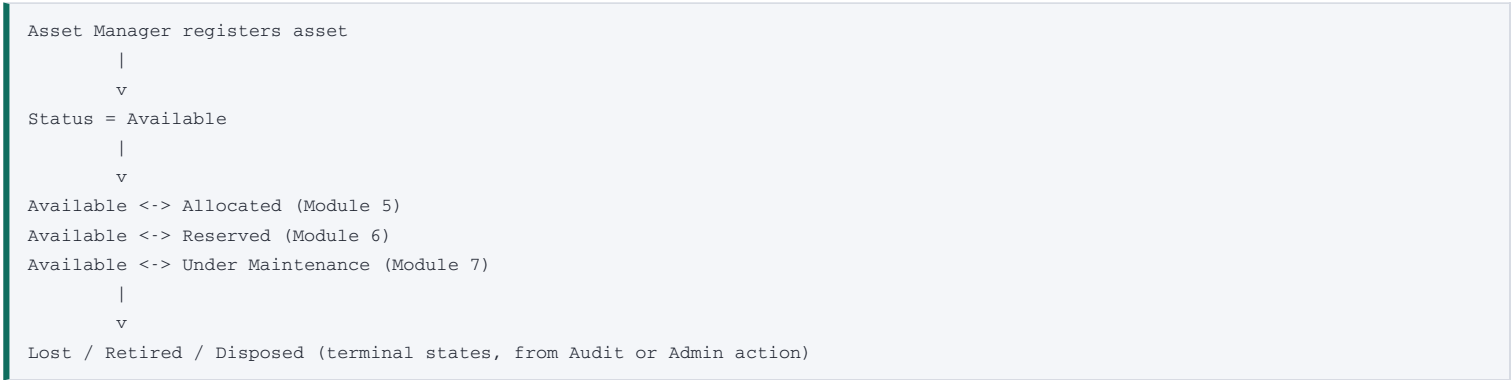
Business Rules

- Asset Tag is system-generated and immutable
- New asset always enters system with Status = Available
- Acquisition Cost is used only for ranking/reports; never posted to accounting
- Status transitions are system-driven, not manually overridden (see state machine below)
- An asset marked Retired/Disposed cannot be allocated or booked

Validations

- Serial Number unique (if provided)
- Category is mandatory
- Acquisition Date cannot be in the future

Workflow



User Roles

Asset Manager: register/edit assets. Admin: full access + retire/dispose. Department Head/Employee: view/search only (scoped).

Dependencies

Depends on Module 3 (Asset Category). Feeds Modules 5, 6, 7, 8, 9.

Module 5 — Asset Allocation & Transfer

Objective

Manage who currently holds each asset, with explicit conflict handling and a controlled transfer workflow.

Features

- Allocate asset to Employee or Department with optional Expected Return Date
- Conflict detection: blocks allocation of an already-allocated asset, shows current holder, offers Transfer Request instead
- Transfer Request workflow: Requested → Approved (Asset Manager/Department Head) → Re-allocated (history auto-updated)
- Return flow: mark returned, capture condition check-in notes, status reverts to Available
- Auto-flagging of overdue allocations (past Expected Return Date) feeding Dashboard + Notifications

Forms

Allocation Form, Transfer Request Form, Return / Check-in Form.

Fields

**Allocation:** Asset (M2O), Allocated To Employee (M2O, optional), Allocated To Department (M2O, optional), Allocation Date, Expected Return Date, Actual Return Date, Status (Active/Returned/Overdue), Condition at Allocation, Condition at Return, Check-in Notes.  
**Transfer Request:** Asset, From Holder, To Holder (Employee/Department), Requested By, Requested Date, Status (Requested/Approved/Rejected/Re-allocated), Approved By, Approval Date, Reason/Notes.

Buttons

Allocate, Request Transfer, Approve Transfer, Reject Transfer, Mark Returned, Save, Cancel.

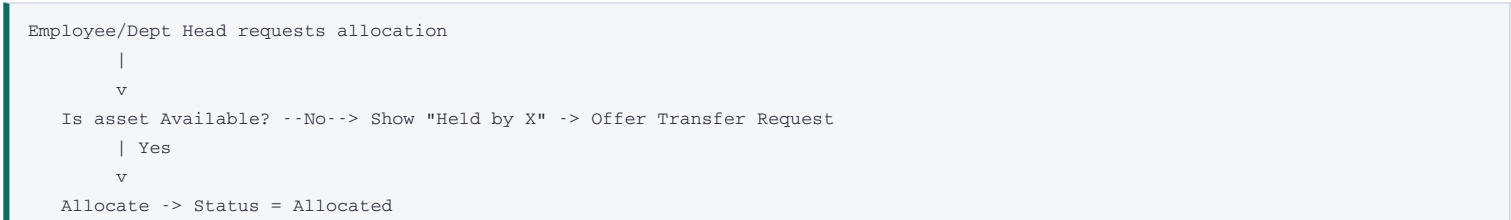
Business Rules

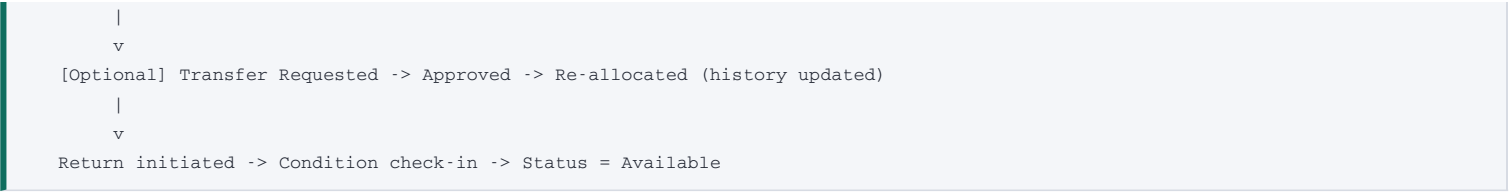
- An asset already Allocated cannot be allocated again — system blocks and shows "currently held by [X]", offering Transfer Request instead
- Transfer Request must be approved by Asset Manager or the relevant Department Head before re-allocation occurs
- On Transfer approval, allocation history is updated automatically (old allocation closed, new allocation opened)
- On Return, asset Status reverts to Available and Actual Return Date is stamped
- Allocation past Expected Return Date is auto-flagged Overdue (scheduled action) and triggers a Notification

Validations

- Expected Return Date, if set, must be on/after Allocation Date
- Asset must be in Available status to be newly allocated
- Transfer Request requires an existing active allocation on the asset

Workflow





User Roles

Asset Manager: allocate, approve transfers, approve returns. Department Head: approve transfers/allocations within department. Employee: initiate return/transfer requests, view own allocations.

Dependencies

Depends on Module 4 (Asset). Feeds Module 2 (Dashboard), Module 10 (Notifications).

Module 6 — Resource Booking

Objective

Enable time-slot booking of shared/bookable resources (rooms, vehicles, equipment) with zero overlaps.

Features

- Calendar view of a resource's existing bookings
- Overlap validation on create/reschedule
- Booking status lifecycle: Upcoming, Ongoing, Completed, Cancelled
- Cancel / Reschedule booking
- Reminder notification before slot starts

Forms

Booking Form, Calendar View, Booking List View.

Fields

Resource/Asset (M2O, shared=True), Booked By (Employee/Department), Start Datetime, End Datetime, Purpose/Notes, Status (Upcoming/Ongoing/Completed/Cancelled), Reminder Sent (boolean).

Buttons

New Booking, Cancel Booking, Reschedule, Save, View Calendar.

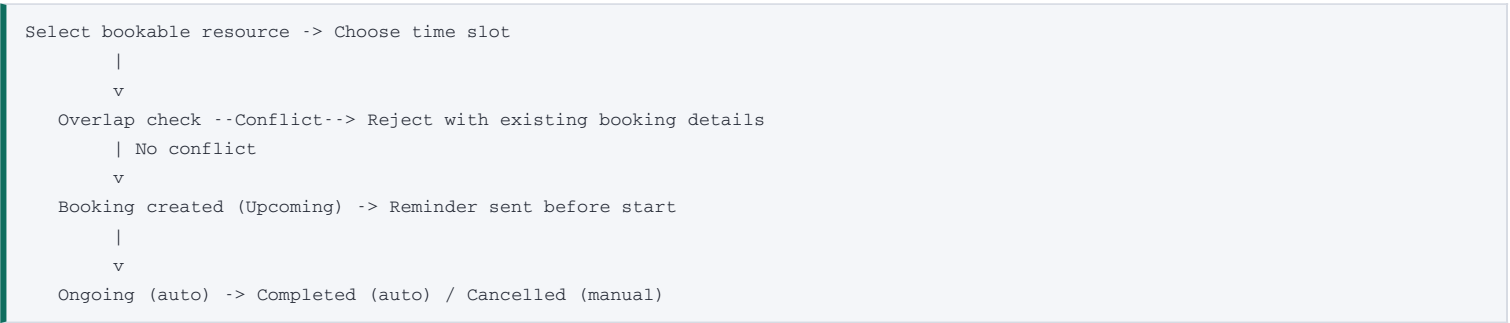
Business Rules

- Only assets flagged "shared/bookable" (Module 4) can be booked
- Two bookings on the same resource cannot have overlapping time ranges; a booking starting exactly when another ends is allowed
- Status auto-transitions by scheduled action: Upcoming → Ongoing → Completed based on current time vs. start/end datetime
- Cancelling a booking releases the slot immediately

Validations

- End Datetime must be after Start Datetime
- Start Datetime cannot be in the past for new bookings
- Overlap check runs against all non-cancelled bookings for that resource

Workflow



User Roles

Employee: book, cancel own bookings. Department Head: book on behalf of department. Asset Manager/Admin: manage all bookings.

Dependencies

Depends on Module 4 (Asset with shared flag). Feeds Module 2, Module 9 (heatmap report), Module 10.

Module 7 — Maintenance Management

Objective

Route asset repair requests through a mandatory approval workflow before work starts.

Features

- Raise maintenance request: asset, issue description, priority, photo attachment
- Approval workflow: Pending → Approved/Rejected (Asset Manager) → Technician Assigned → In Progress → Resolved
- Asset status auto-updates to Under Maintenance on approval, back to Available on resolution
- Maintenance history retained per asset

Forms

Maintenance Request Form, Approval Screen, Technician Assignment Dialog.

Fields

Asset (M2O), Raised By (Employee), Issue Description, Priority (Low/Medium/High/Critical), Photo Attachment, Status (Pending/Approved/Rejected/Technician Assigned/In Progress/Resolved), Approved By, Approval Date, Rejection Reason, Assigned Technician, Resolution Notes, Resolved Date.

Buttons

Raise Request, Approve, Reject, Assign Technician, Start Work, Mark Resolved, Save.

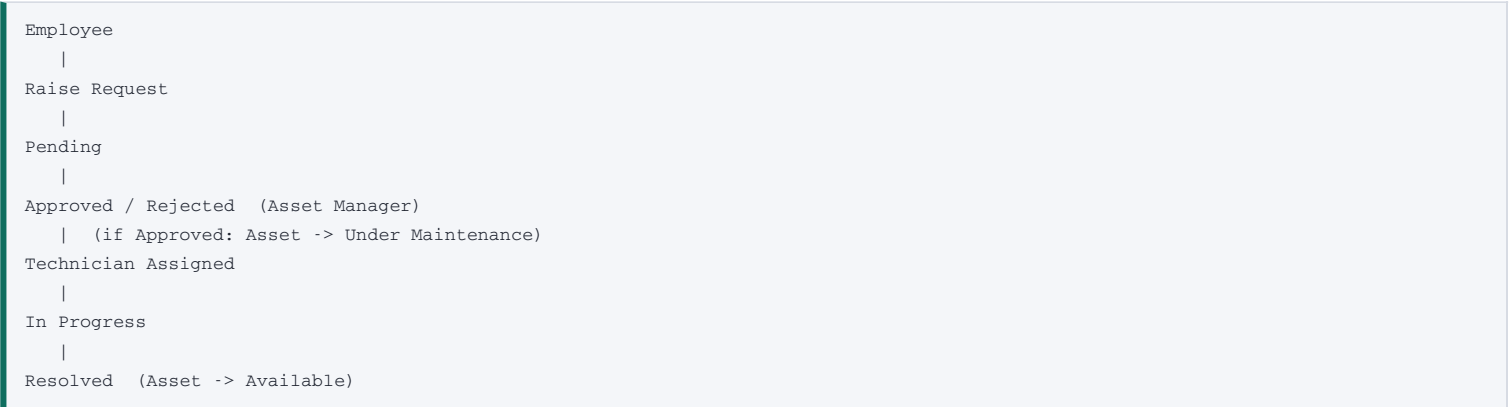
Business Rules

- Maintenance work cannot begin until the request is Approved
- On Approval, linked Asset status is automatically set to Under Maintenance
- On Resolved, linked Asset status is automatically set back to Available
- Rejected requests do not change asset status and require a rejection reason
- An asset can have only one open (non-Resolved/Rejected) maintenance request at a time

Validations

- Issue Description mandatory
- Priority mandatory
- Technician must be assigned before status can move to In Progress

Workflow



User Roles

Employee: raise request (own asset). Asset Manager: approve/reject, assign technician, resolve. Department Head: view department requests. Admin: full visibility.

Dependencies

Depends on Module 4 (Asset). Feeds Module 2, Module 9 (maintenance frequency report), Module 10.

Module 8 — Asset Audit

Objective

Run structured, scheduled verification cycles instead of ad-hoc checks, and auto-generate discrepancy reports.

Features

- Create Audit Cycle: scope (department/location), date range
- Assign one or more auditors to the cycle
- Auditor marks each in-scope asset: Verified / Missing / Damaged
- Auto-generated discrepancy report for flagged items
- Close Audit Cycle: locks the cycle, updates affected asset statuses (e.g. Lost for confirmed-missing)
- Audit history retained per cycle

Forms

Audit Cycle Form, Auditor Assignment, Asset Verification Checklist, Discrepancy Report View.

Fields

**Audit Cycle:** Name, Scope Department/Location, Start Date, End Date, Status (Draft/In Progress/Closed), Assigned Auditors (M2M Employee). **Audit Line:** Audit Cycle (M2O), Asset (M2O), Verified By, Verification Date, Result (Verified/Missing/Damaged), Notes.

Buttons

New Audit Cycle, Assign Auditor, Mark Verified, Mark Missing, Mark Damaged, Generate Discrepancy Report, Close Audit Cycle.

Business Rules

- Only assigned auditors can record verification results for a cycle
- Discrepancy report auto-compiles all Missing/Damaged lines when generated
- Closing a cycle is irreversible: it locks all audit lines and updates asset statuses (Missing → Lost, Damaged → Under Maintenance/flag for review)
- A closed cycle's records cannot be edited

Validations

- End Date must be after Start Date

- At least one auditor must be assigned before cycle can move to In Progress
- Cycle cannot be closed until every in-scope asset has a recorded result

**Workflow**



**User Roles**

Admin: create cycles, close cycles, view all. Asset Manager: assigned as auditor, resolves discrepancies. Department Head: view department audit results.

**Dependencies**

Depends on Module 4 (Asset), Module 3 (Department/Location). Feeds Module 9, Module 10.

**Module 9 — Reports & Analytics**

**Objective**

Give managers actionable, exportable operational insight across assets, bookings, and maintenance.

**Features**

- Asset utilization trends: most-used vs. idle assets
- Maintenance frequency by asset/category
- Assets due for maintenance or nearing retirement
- Department-wise allocation summary
- Resource booking heatmap (peak usage windows)
- Exportable reports (PDF/XLSX/CSV)

**Forms**

Report Filter Panel, Pivot/Graph Views, Export Dialog.

**Fields**

Report Type, Date Range Filter, Department Filter, Category Filter, Output Format (PDF/XLSX/CSV).

**Buttons**

Generate Report, Export PDF, Export Excel, Apply Filters, Reset Filters.

**Business Rules**

- Reports are read-only aggregations; no report action edits transactional data
- Department Head reports are auto-scoped to their department
- Utilization is computed from allocation/booking duration vs. total available time

**Validations**

Date range required for time-bound reports; End Date >= Start Date.

**Workflow**

User selects report type → applies filters → system aggregates data from Modules 4-8 → renders chart/table → optional export.

**User Roles**

Admin/Asset Manager: all reports, org-wide. Department Head: department-scoped reports. Employee: no access (or personal-history view only, configurable).

**Dependencies**

Depends on Modules 4, 5, 6, 7, 8 as data sources.

**Module 10 — Activity Logs & Notifications**

**Objective**

Keep every role informed in real time and maintain a full audit trail of system actions.

**Features**

- Notification triggers: Asset Assigned, Maintenance Approved/Rejected, Booking Confirmed/Cancelled/Reminder, Transfer Approved, Overdue Return Alert, Audit Discrepancy Flagged
- Full activity log of admin/manager/employee actions (who did what, when)
- In-app notification center; optional email notification

**Forms**

Notification Center (list), Activity Log Viewer (filterable table).

**Fields**

**Notification:** Recipient (Employee), Type, Related Record (Reference field), Message, Read Status (boolean), Created Date.

**Activity Log:** User, Action Type, Model/Record Reference, Timestamp, Description/Diff.

**Buttons**

Mark as Read, Mark All as Read, Filter Log, Export Log.

**Business Rules**

- Every state-changing action in Modules 4–8 writes an Activity Log entry automatically
- Notifications are generated by scheduled actions (overdue checks, reminders) and by direct triggers (approvals, confirmations)
- Activity Log is append-only; entries cannot be edited or deleted by any role

**Validations**

N/A (system-generated records).

**Workflow**

Triggering event occurs in any module → system creates Notification record(s) for relevant recipient(s) and an Activity Log entry → recipient sees badge/alert on next screen load or via email/cron digest.

**User Roles**

All roles receive scoped notifications. Admin: full activity log visibility. Others: own/department-scoped log visibility.

**Dependencies**

Depends on all other modules as event sources.

### 3. Screen-wise Functional Requirements

#### Login / Signup Screen

| Screen          | Functional Requirement                                       | Priority | Assigned Module |
|-----------------|--------------------------------------------------------------|----------|-----------------|
| Login           | Email & password authentication with session validation      | High     | Module 1        |
| Signup          | Self-signup creates Employee account only, no role selection | High     | Module 1        |
| Forgot Password | Reset password via emailed link                              | Medium   | Module 1        |

#### Dashboard

| Screen    | Functional Requirement                                                                                    | Priority | Assigned Module |
|-----------|-----------------------------------------------------------------------------------------------------------|----------|-----------------|
| Dashboard | KPI cards (Available, Allocated, Maintenance Today, Active Bookings, Pending Transfers, Upcoming Returns) | High     | Module 2        |
| Dashboard | Overdue returns highlighted separately from upcoming returns                                              | High     | Module 2        |
| Dashboard | Quick actions: Register Asset, Book Resource, Raise Maintenance Request                                   | Medium   | Module 2        |

#### Organization Setup

| Screen                 | Functional Requirement                                           | Priority | Assigned Module |
|------------------------|------------------------------------------------------------------|----------|-----------------|
| Department Tab         | Create/edit/deactivate department with head and parent hierarchy | High     | Module 3        |
| Asset Category Tab     | Create/edit categories with optional custom fields               | High     | Module 3        |
| Employee Directory Tab | List employees; promote to Department Head/Asset Manager         | High     | Module 3        |

#### Asset Registration

| Screen             | Functional Requirement                                                   | Priority | Assigned Module |
|--------------------|--------------------------------------------------------------------------|----------|-----------------|
| Asset Registration | Register asset with auto Asset Tag, QR, category, condition, photo       | High     | Module 4        |
| Asset Directory    | Search/filter by tag, serial, QR, category, status, department, location | High     | Module 4        |
| Asset Detail       | Show lifecycle status and allocation + maintenance history               | Medium   | Module 4        |

#### Asset Allocation

| Screen           | Functional Requirement                                    | Priority | Assigned Module |
|------------------|-----------------------------------------------------------|----------|-----------------|
| Allocation       | Allocate asset to employee/department with conflict check | High     | Module 5        |
| Transfer Request | Requested → Approved → Re-allocated workflow              | High     | Module 5        |
| Return           | Mark returned with condition check-in notes               | Medium   | Module 5        |

#### Booking

| Screen           | Functional Requirement                       | Priority | Assigned Module |
|------------------|----------------------------------------------|----------|-----------------|
| Booking Calendar | Calendar view of resource bookings           | High     | Module 6        |
| New Booking      | Overlap validation on create                 | High     | Module 6        |
| Booking List     | Cancel/reschedule with reminder notification | Medium   | Module 6        |

#### Maintenance

| Screen              | Functional Requirement                                    | Priority | Assigned Module |
|---------------------|-----------------------------------------------------------|----------|-----------------|
| Raise Request       | Select asset, describe issue, set priority, attach photo  | High     | Module 7        |
| Approval            | Approve/Reject by Asset Manager; auto asset status update | High     | Module 7        |
| Technician Workflow | Assign → In Progress → Resolved                           | Medium   | Module 7        |

#### Audit

| Screen                 | Functional Requirement                                 | Priority | Assigned Module |
|------------------------|--------------------------------------------------------|----------|-----------------|
| Audit Cycle Setup      | Create cycle with scope, date range, assigned auditors | High     | Module 8        |
| Verification Checklist | Mark each asset Verified/Missing/Damaged               | High     | Module 8        |
| Discrepancy Report     | Auto-generate report for flagged items                 | High     | Module 8        |
| Close Cycle            | Lock records, update asset statuses                    | Medium   | Module 8        |

#### Reports

| Screen             | Functional Requirement      | Priority | Assigned Module |
|--------------------|-----------------------------|----------|-----------------|
| Utilization Report | Most-used vs. idle assets   | Medium   | Module 9        |
| Maintenance Report | Frequency by asset/category | Medium   | Module 9        |

|                    |                                    |     |          |
|--------------------|------------------------------------|-----|----------|
| Allocation Summary | Department-wise allocation summary | Low | Module 9 |
| Booking Heatmap    | Peak usage windows visualization   | Low | Module 9 |

Notifications

| Screen              | Functional Requirement                                   | Priority | Assigned Module |
|---------------------|----------------------------------------------------------|----------|-----------------|
| Notification Center | List of role-scoped notifications with read/unread state | Medium   | Module 10       |
| Activity Log        | Full filterable log of system actions                    | Low      | Module 10       |

## 4. Database Entities

### Department

|                    |                                                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Fields             | id, name, code, parent_department_id, department_head_id, status                                                            |
| Relationships      | M2O parent_department_id → Department (self); M2O department_head_id → Employee; O2M to Employee, Asset Allocation, Booking |
| Mandatory Fields   | name, status                                                                                                                |
| Unique Constraints | name (per company), code                                                                                                    |

### Asset Category

|                    |                                                                 |
|--------------------|-----------------------------------------------------------------|
| Fields             | id, name, code, parent_category_id, custom_field_config, status |
| Relationships      | M2O parent_category_id → Asset Category (self); O2M to Asset    |
| Mandatory Fields   | name, status                                                    |
| Unique Constraints | name                                                            |

### Employee

|                    |                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------|
| Fields             | id, name, email, department_id, role, status, user_id (linked login)                                            |
| Relationships      | M2O department_id → Department; O2M to Asset Allocation, Booking, Maintenance Request, Audit Line, Notification |
| Mandatory Fields   | name, email, role, status                                                                                       |
| Unique Constraints | email                                                                                                           |

### Asset

|                    |                                                                                                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fields             | id, asset_tag (auto), name, category_id, serial_number, acquisition_date, acquisition_cost, condition, location, shared_flag, qr_code, status, current_holder_id (computed) |
| Relationships      | M2O category_id → Asset Category; O2M to Allocation, Booking, Maintenance Request, Audit Line                                                                               |
| Mandatory Fields   | name, category_id, status                                                                                                                                                   |
| Unique Constraints | asset_tag, serial_number (if provided)                                                                                                                                      |

### Allocation

|                    |                                                                                                                                                                          |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fields             | id, asset_id, employee_id, department_id, allocation_date, expected_return_date, actual_return_date, status, condition_at_allocation, condition_at_return, checkin_notes |
| Relationships      | M2O asset_id → Asset; M2O employee_id → Employee; M2O department_id → Department                                                                                         |
| Mandatory Fields   | asset_id, allocation_date, status                                                                                                                                        |
| Unique Constraints | Only one Active allocation per asset (enforced in business logic)                                                                                                        |

### Transfer Request

|                    |                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------|
| Fields             | id, asset_id, from_holder, to_holder, requested_by, requested_date, status, approved_by, approval_date, reason |
| Relationships      | M2O asset_id → Asset; M2O requested_by/approved_by → Employee                                                  |
| Mandatory Fields   | asset_id, to_holder, requested_by, status                                                                      |
| Unique Constraints | None (multiple historical transfers per asset allowed)                                                         |

### Booking

|                    |                                                                                          |
|--------------------|------------------------------------------------------------------------------------------|
| Fields             | id, asset_id, booked_by_id, start_datetime, end_datetime, purpose, status, reminder_sent |
| Relationships      | M2O asset_id → Asset (shared=True); M2O booked_by_id → Employee/Department               |
| Mandatory Fields   | asset_id, start_datetime, end_datetime, status                                           |
| Unique Constraints | No overlapping (asset_id, start_datetime, end_datetime) for non-cancelled bookings       |

### Maintenance Request

|                    |                                                                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fields             | id, asset_id, raised_by_id, issue_description, priority, photo, status, approved_by, approval_date, rejection_reason, technician_id, resolution_notes, resolved_date |
| Relationships      | M2O asset_id → Asset; M2O raised_by_id/approved_by/technician_id → Employee                                                                                          |
| Mandatory Fields   | asset_id, issue_description, priority, status                                                                                                                        |
| Unique Constraints | Only one open request per asset (business logic)                                                                                                                     |

### Audit Cycle

|                    |                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------|
| Fields             | id, name, scope_department_id, scope_location, start_date, end_date, status, auditor_ids (M2M) |
| Relationships      | M2O scope_department_id → Department; M2M auditor_ids → Employee; O2M to Audit Line            |
| Mandatory Fields   | name, start_date, end_date, status                                                             |
| Unique Constraints |                                                                                                |

|                    |      |
|--------------------|------|
| Unique Constraints | None |
|--------------------|------|

Audit Line (Discrepancy Source)

|                    |                                                                                       |
|--------------------|---------------------------------------------------------------------------------------|
| Fields             | id, audit_cycle_id, asset_id, verified_by_id, verification_date, result, notes        |
| Relationships      | M2O audit_cycle_id → Audit Cycle; M2O asset_id → Asset; M2O verified_by_id → Employee |
| Mandatory Fields   | audit_cycle_id, asset_id, result                                                      |
| Unique Constraints | (audit_cycle_id, asset_id) unique together                                            |

Notification

|                    |                                                                                |
|--------------------|--------------------------------------------------------------------------------|
| Fields             | id, recipient_id, type, related_record_ref, message, read_status, created_date |
| Relationships      | M2O recipient_id → Employee; Reference field to any source record              |
| Mandatory Fields   | recipient_id, type, message                                                    |
| Unique Constraints | None                                                                           |

Activity Log

|                    |                                                                        |
|--------------------|------------------------------------------------------------------------|
| Fields             | id, user_id, action_type, model_ref, record_id, timestamp, description |
| Relationships      | M2O user_id → Employee; Reference field to source model/record         |
| Mandatory Fields   | user_id, action_type, timestamp                                        |
| Unique Constraints | None (append-only log)                                                 |

## 5. Business Rules

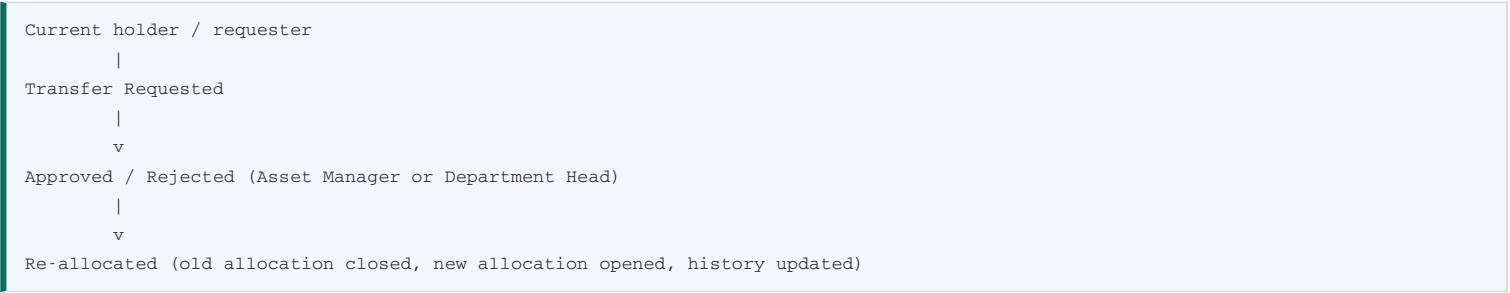
- Signup creates an Employee account only — no role selection at signup; roles are assigned solely by Admin from the Employee Directory
- An asset cannot be allocated twice — a currently-allocated asset shows "currently held by [X]" and offers Transfer Request instead of allocation
- Booking overlap is not allowed on the same resource; a booking starting exactly when a prior one ends is permitted
- Only Admin assigns/promotes roles (Department Head, Asset Manager)
- Maintenance work requires Approval before it can begin; approval auto-flips asset status to Under Maintenance
- Asset status updates automatically based on workflow events (allocation, return, maintenance approval/resolution, audit closure) — never manually overridden
- Audit Cycle closure locks all audit lines and is irreversible
- Overdue allocations, bookings, and maintenance trigger notifications automatically via scheduled actions
- Transfer Requests must be approved by Asset Manager or the relevant Department Head before re-allocation
- A newly registered asset always enters the system with status Available
- Acquisition Cost is used only for reporting/ranking and is never linked to accounting/invoicing
- An asset with status Retired or Disposed cannot be allocated or booked
- Only assets flagged "shared/bookable" appear in the Resource Booking module
- An asset can have only one open (non-terminal) Maintenance Request at a time
- Activity Log entries are system-generated and append-only; no role can edit or delete them

## 6. Workflows

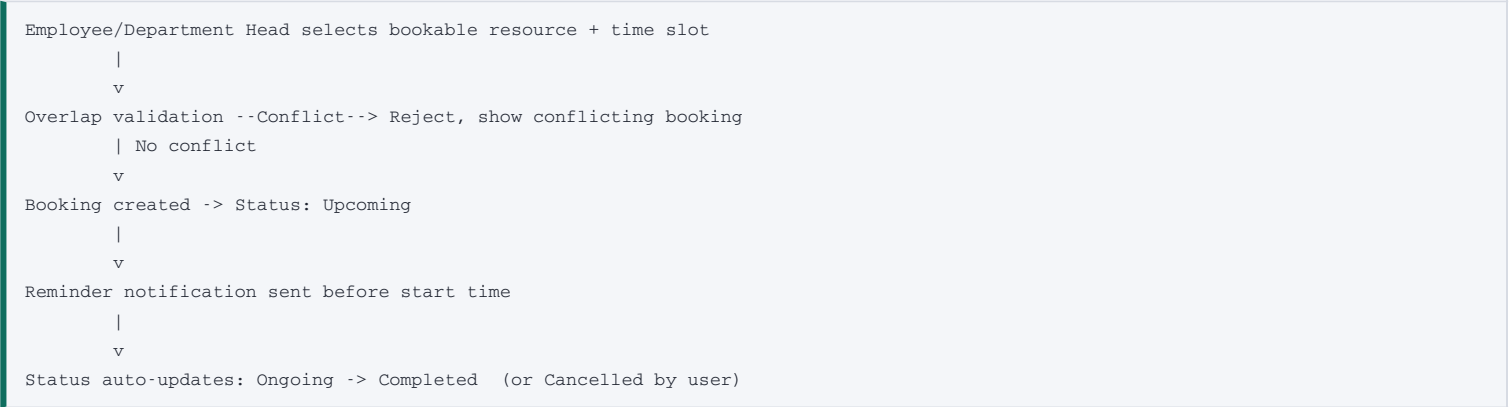
### Asset Allocation



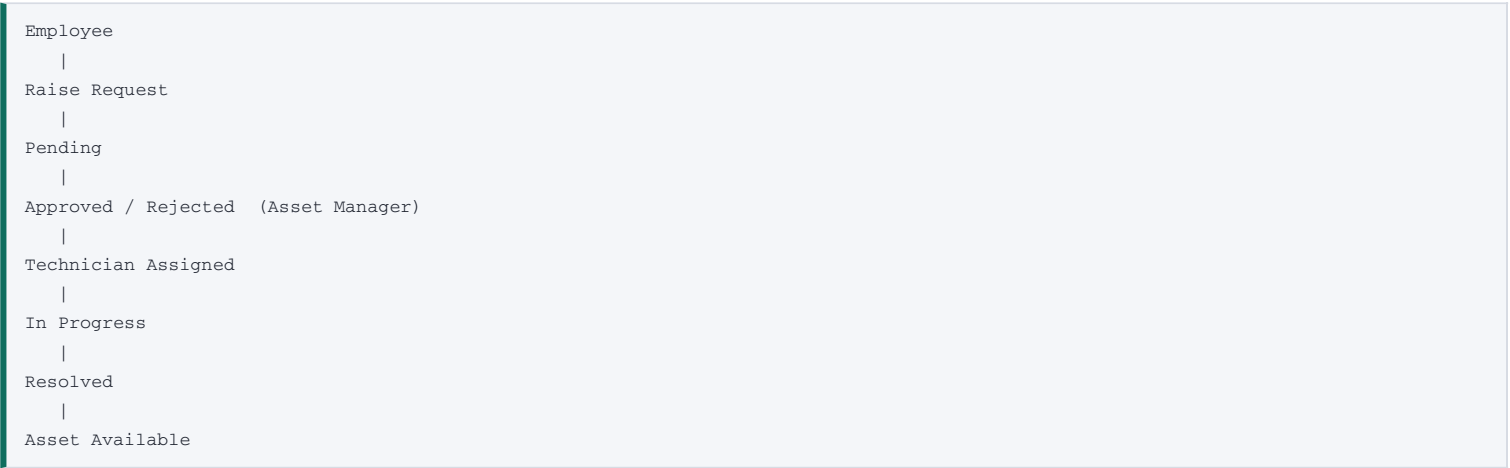
### Transfer



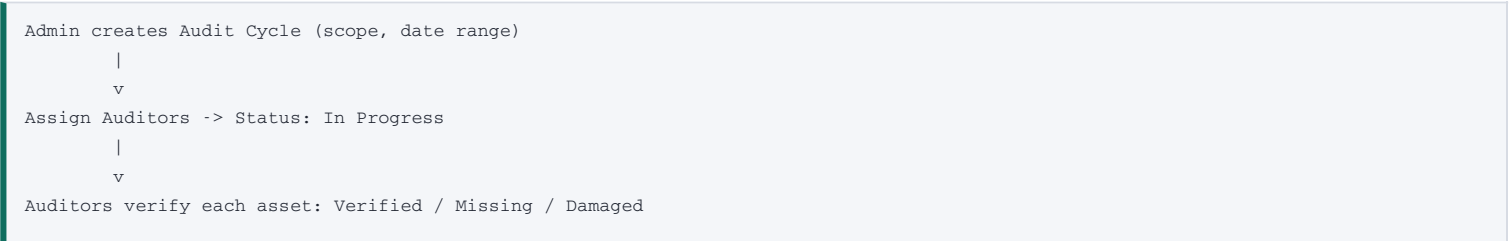
### Booking



### Maintenance



### Audit



|

v

Discrepancy Report auto-generated for flagged items

|

v

Close Audit Cycle -> Locks records -> Updates asset status (e.g. Lost)

## 7. User Roles & Permissions Matrix

| Feature                                  | Admin      | Asset Manager                | Department Head      | Employee            |
|------------------------------------------|------------|------------------------------|----------------------|---------------------|
| Organization Setup (Dept/Category/Roles) | CRUD       | —                            | —                    | —                   |
| Employee Directory (view)                | CRUD       | Read                         | Read (dept)          | Read (self)         |
| Asset Registration                       | CRUD       | CRUD                         | Read                 | Read                |
| Asset Search/Directory                   | Read       | Read                         | Read (dept)          | Read (own)          |
| Asset Allocation                         | CRUD       | Create/Update                | Approve (dept)       | Request only        |
| Transfer Request                         | CRUD       | Approve                      | Approve (dept)       | Create (own)        |
| Resource Booking                         | CRUD       | CRUD                         | Create/Update (dept) | Create/Update (own) |
| Maintenance Request                      | CRUD       | Approve/Assign/Resolve       | Read (dept)          | Create (own)        |
| Audit Cycle Setup                        | CRUD       | —                            | —                    | —                   |
| Audit Verification                       | Read       | Update (if assigned auditor) | Read (dept)          | —                   |
| Reports & Analytics                      | Read (all) | Read (all)                   | Read (dept)          | —                   |
| Notifications                            | Read (all) | Read (own)                   | Read (own)           | Read (own)          |
| Activity Log                             | Read (all) | Read (own actions)           | Read (dept)          | Read (own actions)  |

CRUD = Create, Read, Update, Delete. Scope qualifiers (dept/own/all) map directly to Odoo Record Rules.

## 8. Odoo Module Mapping

| Odoo Concept                        | Application in AssetFlow                                                                                                                                                                                                                                                                                               |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security (Groups)                   | assetflow.group_admin, group_asset_manager, group_department_head, group_employee — mapped to res.groups with inherited base.group_user                                                                                                                                                                                |
| Access Rights (ir.model.access.csv) | Per-model CRUD matrix aligned to Section 7 permissions matrix                                                                                                                                                                                                                                                          |
| Record Rules (ir.rule)              | Department-scoped and self-scoped domains, e.g. Department Head sees allocation.department_id = user's department; Employee sees allocation.employee_id = user                                                                                                                                                         |
| Models                              | assetflow.department, asset.category, hr.employee (extended) or assetflow.employee, assetflow.asset, assetflow.allocation, assetflow.transfer.request, assetflow.booking, assetflow.maintenance.request, assetflow.audit.cycle, assetflow.audit.line, assetflow.notification (or mail.message), assetflow.activity.log |
| Views                               | Form, Tree/List, Kanban (Dashboard KPIs), Calendar (Booking), Pivot/Graph (Reports), Search views with filters/group-by                                                                                                                                                                                                |
| Menus                               | Top-level "AssetFlow" app menu with sub-menus per module (Setup, Assets, Allocation, Booking, Maintenance, Audit, Reports, Notifications)                                                                                                                                                                              |
| Actions (ir.actions.act_window)     | One window action per menu item; contextual actions for Transfer Request, Approve, Close Cycle                                                                                                                                                                                                                         |
| Sequences (ir.sequence)             | Auto Asset Tag generation (AF-0001...), Audit Cycle reference numbers                                                                                                                                                                                                                                                  |
| Mail Thread / Chatter (mail.thread) | Enabled on Asset, Maintenance Request, Transfer Request, Audit Cycle for activity tracking and internal notes                                                                                                                                                                                                          |
| Activities (mail.activity)          | Follow-ups for pending approvals (transfer, maintenance) assigned to Asset Manager/Department Head                                                                                                                                                                                                                     |
| Scheduled Actions (ir.cron)         | Overdue allocation/booking check, booking status auto-transition (Upcoming→Ongoing→Completed), reminder notifications                                                                                                                                                                                                  |
| Reports (QWeb / ir.actions.report)  | Asset Utilization, Maintenance Frequency, Department Allocation Summary, Discrepancy Report — PDF/XLSX export                                                                                                                                                                                                          |
| Dashboards                          | Odoo Dashboard/Spreadsheet or custom OWL Kanban with KPI tiles bound to computed fields                                                                                                                                                                                                                                |

## 9. Development Task Breakdown

### Team Member 1 — Authentication, Security & Organization Setup

| Task                                                   | Est. Hours | Dependencies            | Priority | Complexity |
|--------------------------------------------------------|------------|-------------------------|----------|------------|
| Authentication (signup/login/reset), Security Groups   | 1.5        | None                    | High     | Medium     |
| Employee model & Directory screen                      | 1          | Authentication          | High     | Low        |
| Department model & Setup Tab A                         | 1          | Employee model          | High     | Low        |
| Role promotion logic (Admin-only)                      | 1          | Employee, Department    | High     | Medium     |
| Access Rights & Record Rules (all models, coordinated) | 1.5        | All team models defined | High     | High       |
| Menus & App structure                                  | 0.5        | Above                   | Medium   | Low        |

### Team Member 2 — Asset Category, Registration & Directory

| Task                                              | Est. Hours | Dependencies                              | Priority | Complexity |
|---------------------------------------------------|------------|-------------------------------------------|----------|------------|
| Asset Category model & Setup Tab B                | 1          | Department module skeleton                | High     | Low        |
| Asset model, sequence-based Asset Tag             | 1.5        | Asset Category                            | High     | Medium     |
| Asset Registration form (photo, docs, condition)  | 1          | Asset model                               | High     | Low        |
| QR code generation & scan lookup                  | 1          | Asset model                               | Medium   | Medium     |
| Search/filter Directory view + Asset History tabs | 1          | Allocation, Maintenance models (stubs ok) | High     | Medium     |
| Lifecycle status state machine                    | 0.5        | Asset model                               | High     | Medium     |

### Team Member 3 — Allocation, Transfer, Booking, Dashboard & Notifications

| Task                                               | Est. Hours | Dependencies                           | Priority | Complexity |
|----------------------------------------------------|------------|----------------------------------------|----------|------------|
| Allocation model + conflict-check logic            | 1.5        | Asset model                            | High     | High       |
| Transfer Request workflow                          | 1          | Allocation                             | High     | Medium     |
| Return / condition check-in flow                   | 0.5        | Allocation                             | Medium   | Low        |
| Booking model + overlap validation + calendar view | 1.5        | Asset model (shared flag)              | High     | High       |
| Dashboard KPI widgets                              | 1          | Allocation, Booking, Maintenance stubs | High     | Medium     |
| Notification model + basic triggers                | 1          | All transactional models               | Medium   | Medium     |

### Team Member 4 — Maintenance, Audit, Reports & Analytics

| Task                                                                             | Est. Hours | Dependencies                          | Priority | Complexity |
|----------------------------------------------------------------------------------|------------|---------------------------------------|----------|------------|
| Maintenance Request model + approval workflow                                    | 1.5        | Asset model                           | High     | Medium     |
| Asset status auto-update on approve/resolve                                      | 0.5        | Maintenance Request                   | High     | Medium     |
| Audit Cycle + Audit Line models                                                  | 1.5        | Asset, Department models              | High     | High       |
| Discrepancy report + Close Cycle logic                                           | 1          | Audit Cycle/Line                      | High     | Medium     |
| Reports: utilization, maintenance frequency, allocation summary, booking heatmap | 1.5        | Allocation, Booking, Maintenance data | Medium   | High       |
| Export (PDF/XLSX) wiring                                                         | 0.5        | Reports                               | Low      | Low        |

# 10. One-Day Hackathon Plan (10:00 AM - 5:00 PM)

| Time          | Activity                                                                                                   | Owner(s)                       |
|---------------|------------------------------------------------------------------------------------------------------------|--------------------------------|
| 10:00 - 10:30 | Project setup: repo, Odoo module scaffold, shared model naming conventions, security groups skeleton       | All (TM1 leads)                |
| 10:30 - 11:30 | Core models: Department, Category, Employee, Asset, Allocation, Booking, Maintenance, Audit Cycle          | All (parallel, per assignment) |
| 11:30 - 1:00  | Views: forms, list, search, calendar, kanban dashboard; menus and actions wired                            | All (parallel)                 |
| 1:00 - 2:00   | Lunch / Integration checkpoint — merge branches, resolve model dependency conflicts                        | All                            |
| 2:00 - 3:30   | Business logic: conflict checks, overlap validation, approval workflows, state machines, scheduled actions | All (parallel)                 |
| 3:30 - 4:30   | Testing: cross-module flows (allocate→transfer→return, book→overlap, raise→approve→resolve, audit→close)   | All                            |
| 4:30 - 5:00   | Bug fixes, demo data seeding, final walkthrough rehearsal                                                  | All                            |

## 11. Deliverables Checklist

- ☐ Authentication module (signup/login/reset, no self-elevation)
- ☐ Security groups & access rights configured for all 4 roles
- ☐ Menus structured under a single AssetFlow app
- ☐ Form, List, Search, Calendar, and Kanban views for all modules
- ☐ All 12 core models implemented with required fields/relations
- ☐ Reports: utilization, maintenance frequency, allocation summary, booking heatmap (PDF/XLSX export)
- ☐ Access Rights (ir.model.access.csv) and Record Rules (ir.rule) per role scope
- ☐ Notification triggers for the 7 listed event types
- ☐ Scheduled Actions (Cron Jobs): overdue check, booking auto-status transition, reminders
- ☐ Dashboard with KPI cards and quick actions
- ☐ End-to-end testing of Allocation, Transfer, Booking, Maintenance, and Audit workflows

## 12. Out of Scope

- Purchasing, procurement, and vendor management
- Invoicing, billing, or any accounting/financial ledger integration (Acquisition Cost is reporting-only)
- Payroll or HR functions beyond the basic Employee directory required for role/department context
- Mobile native applications (only responsive web UI is in scope)
- Third-party integrations not named in the problem statement (e.g. IoT asset tracking, GPS telemetry)
- Multi-company / multi-currency configuration
- Advanced BI/analytics beyond the reports explicitly listed in Section 9 (Reports Screen)